

# FLAVOURBENCH: Ranking Frontier Language Models with Executable Culinary Ground Truth

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## Abstract

Open-ended language-model benchmarks usually inherit a judge: a human preference panel, another model, or a brittle exact-match key. We introduce FLAVOURBENCH, an automated benchmark in which a versioned culinary system supplies dense, executable ground truth. Each task presents eight ingredients and asks for a three-ingredient portfolio; before model execution, EPICURE scores all 56 possible portfolios. We evaluate 27 frontier endpoints on an identical 534-task core spanning substitution, pairing, and constrained composition. Every ranked model has exactly 89 valid responses per panel and family (14418 model-task cells total), eliminating differential missingness from the leaderboard. The FlavourBench Score is the equal-family mean of the frozen task scores. We use 50000 anchor-cluster bootstrap replicates for simultaneous 95% score bands and 100000 sign-flip draws for all 351 paired model contrasts, with Holm control. The two independently compiled panels correlate at  $r = 0.89$  (rank  $\rho = 0.80$ ). Grok 4.6 has the largest point estimate at 65.1 (simultaneous 95% CI 61.0–69.2); 101 of 351 model pairs are resolved. The release includes the prompts, all portfolio score maps, raw responses, exact routes, content hashes, and an offline verifier that reconstructs every result.

## 1 Introduction

Benchmarks are strongest when the evaluator can be executed. Code can be tested, database states can be inspected, and games can be scored. Culinary reasoning is usually evaluated with a model judge, a handful of human preferences, or factual questions whose answers do not capture the quality of a complete decision. That leaves two problems: the judge is entangled with the systems being tested, and exact match throws away useful differences between plausible answers.

FLAVOURBENCH turns a culinary runtime into a benchmark environment. EPICURE represents 1,790 ingredients in a 300-dimensional space and exposes deterministic operations for substitution, pairing, dietary feasibility, and regional composition [12, 13]. A task compiler converts these operations into three-of-eight selection problems. It enumerates every candidate portfolio and freezes a continuous 0–100 score map before model execution. An answer is therefore scored by table lookup, without a model judge and without post-hoc interpretation.

This setup separates the reference system from the evaluated systems. EPICURE is not a leaderboard entry and is not “100% accurate.” It defines the task-specific reward surface. A model receives 100 on a task only if it selects that task’s highest-scoring portfolio; other valid portfolios receive partial credit. The benchmark measures agreement with a published executable culinary environment, not universal human taste.

We use that environment to evaluate 27 current endpoints. Every model faces the same 534 tasks and 56 frozen answer scores per task. The complete common core supports direct paired comparisons on 534 shared

observations per model pair instead of inferring a ranking from sparse votes or comparing endpoints on different task subsets.

The paper makes three contributions:

1. an executable culinary benchmark with 29904 model-independent portfolio scores and a single interpretable 0–100 metric;
2. a 27-model, 534-task complete-core evaluation with simultaneous uncertainty and paired, multiplicity-controlled tests; and
3. a content-addressed release in which every public number traces to frozen tasks, exact routes, raw responses, and a dependency-light offline scorer.

## 2 Executable culinary ground truth

### 2.1 A reference environment, not a model judge

The benchmark binds an exact EPICURE release, evidence bundle, application build, ingredient count, and embedding dimensionality by hash. Changing any of these inputs creates a new benchmark task-set identity. This is analogous to fixing a simulator or a code-test suite: the reference is explicit and reproducible, but its external validity remains a scientific question rather than an assumption hidden inside a judge prompt.

For each task  $t$ , let  $\mathcal{A}_t$  contain all 56 three-item subsets of eight candidates. EPICURE assigns a raw utility  $u_t(a)$  to every valid  $a \in \mathcal{A}_t$ . Invalid portfolios under a dietary or category constraint receive zero. Valid utilities are

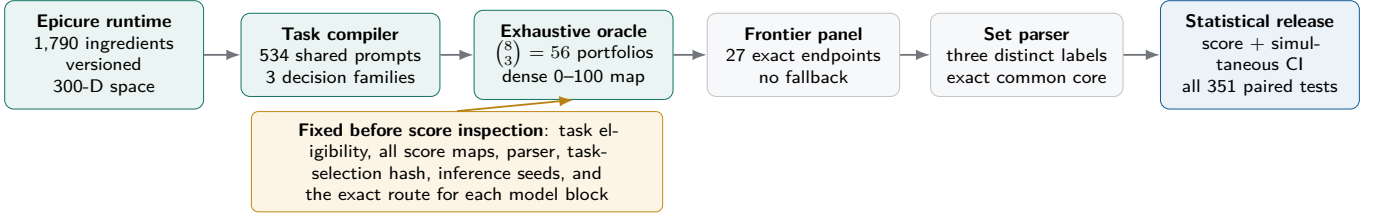


Figure 1: FLAVOURBENCH compiles a versioned culinary runtime into dense selection tasks. The evaluated models never judge one another and do not access EPICURE. All score maps and the per-panel contracts are fixed before collection. The complete-core task set is selected from response validity alone, before inspecting any model selection or score.

normalized within the task:

$$s_t(a) = 100 \frac{u_t(a) - \min_{b \in \mathcal{A}_t} u_t(b)}{\max_{b \in \mathcal{A}_t} u_t(b) - \min_{b \in \mathcal{A}_t} u_t(b)}. \quad (1)$$

Every task has a unique 100-point optimum. The exact chance baseline is not a generic percentage; it is  $|\mathcal{A}_t|^{-1} \sum_a s_t(a)$  for that frozen task.

## 2.2 Three complete-core decision families

The ranked core contains 534 tasks: 89 from each family in each of two independently compiled panels, spanning 534 unique anchor ingredients. Tasks sharing an anchor move together as one cluster in inference. Candidate sets are selected across validation strata, then ordered and labeled by a task-derived hash. No evaluated model writes an item, distractor, or score. Regional-composition tasks are released as a supplemental track but are not in the 27-model leaderboard because they do not support an equally complete common core.

The score maps have substantial resolution rather than one correct key plus 55 equivalent mistakes. Table 2 reports the exact random-portfolio baseline, separation of the best and second-best portfolios, and the number of distinct score values. Constraint tasks are intentionally sparse because infeasible portfolios score zero; the other families provide much finer graded rewards.

## 2.3 Prompt and parser contract

Prompts show the decision context and eight labeled candidates. The final line must begin with the `FINAL_SELECTION` marker and contain three distinct A–H labels separated by commas. No prompt supplies a concrete valid triplet. The parser takes the final exact marker line, uppercases the labels, sorts them as an unordered set, and performs one lookup in the frozen score map. Explanatory text does not affect the result. Missing markers, duplicate labels, provider failures, and abnormal finishes remain in the raw response ledger but cannot enter the common core.

The model-only condition is deliberate. EPICURE supplies the ground truth but is not exposed as a tool during ranking. Thus the score measures whether a model can

make the culinary decision, not whether it can copy a runtime return value after being told which operation to call.

## 3 Evaluation design

### 3.1 One primary score

For model  $m$ , panel  $p$ , family  $f$ , and task  $t$ , let  $Y_{mt}$  be the frozen portfolio score and let  $T_{pf}$  be the score-blindly selected set of 89 tasks completed by every model. The FlavourBench Score is

$$F_m = \frac{1}{2 \cdot 3} \sum_{p=1}^2 \sum_{f=1}^3 \frac{1}{|T_{pf}|} \sum_{t \in T_{pf}} Y_{mt}. \quad (2)$$

It ranges from 0 to 100 and equals the ordinary mean over all 534 cells because the six panel–family strata have equal size. Every ranked endpoint contributes exactly 534 valid responses. Failures are neither assigned zero nor silently dropped model by model: they determine whether a task can enter the shared core, using response status and parser validity only. The task-selection hash is fixed before loading any selected portfolio or score.

### 3.2 Frontier endpoint panel

The panel spans OpenAI, Anthropic, Google, xAI, Meta, Kimi, Qwen, Z.ai, DeepSeek, ByteDance, Thinking Machines, MiniMax, NVIDIA, Mistral, Tencent, and Cohere. It includes GPT-5.6 Sol, Terra, and Luna; Claude Opus, Sonnet, and Fable 5; Gemini 3.1 Pro and 3.6 Flash; Grok 4.6; Llama 4 Maverick; Muse Spark 1.2 and Muse Glimmer 30B; Kimi K3; Qwen 3.8 Max and Qwen3.8 2.4T A95B; GLM 5.2 and GLM 5.3; DeepSeek V4 Pro 0813 and V4 Flash; Seed 2.1 Turbo; Inkling; MiniMax M3; Nemotron 3.5 Lightning; Mistral Large 3; Tencent HY 3; Cohere Command A; and Cohere Command R+ (08-2024). Z.ai supplied written permission for one finite GLM 5.3 benchmark run, conditioned on the adapter not becoming a permanent running function. We bind that permission and its hash in the release, require the returned identity to equal `glm-5.3`, and expose no standing endpoint.<sup>1</sup>

<sup>1</sup><https://docs.z.ai/guides/llm/glm-5.3>

| Family       | Decision                                                           | Epicure utility                                | Hard constraint             | Tasks |
|--------------|--------------------------------------------------------------------|------------------------------------------------|-----------------------------|-------|
| Substitution | choose a three-item replacement portfolio for an anchor ingredient | 0.8 anchor similarity +0.2 portfolio coherence | none                        | 178   |
| Pairing      | choose three ingredients to accompany an anchor                    | 0.65 anchor affinity +0.35 portfolio coherence | none                        | 178   |
| Constraints  | choose a feasible portfolio under diet and processing limits       | 0.7 anchor similarity +0.3 coherence           | diet and maximum NOVA level | 178   |

Table 1: The three complete-core task families share one response and scoring interface but probe different culinary decisions. Each task contains eight candidates and 56 frozen scores.

| Family       | Exact chance | Top gap | Distinct scores |
|--------------|--------------|---------|-----------------|
| Substitution | 45.2         | 5.6     | 56              |
| Pairing      | 45.0         | 5.6     | 56              |
| Constraints  | 3.4          | 37.3    | 4               |

Table 2: Frozen task-map diagnostics for 178 scheduled tasks per family, computed before model execution. Chance is the uniformly random portfolio mean; top gap is the median best-second-best margin.

Routes were chosen before each scored block and automatic fallback is disabled. Most endpoints use an exact OpenRouter route; Qwen 3.8 Max uses Alibaba, Fable 5 uses Google Vertex, GLM 5.3 uses the finite Z.ai route above, and selected DeepSeek blocks use an exact GMI-Cloud or DeepInfra route. When transport validation required a replacement, the complete model block was rerun under one new route; successful cells from superseded blocks are not pooled. The manifest binds the requested model, returned identity contract, endpoint tag, provider, supported parameters, and execution contract for both panels. The exact per-model routes appear in Appendix A. For endpoints that advertise a reasoning-effort control, hidden reasoning is excluded and effort is fixed to the lowest transport-stable setting recorded in the exact analysis plan. Endpoints without that control use their provider-fixed behavior. This prevents hidden deliberation from silently consuming the answer budget while leaving the scored response format identical across routes.

### 3.3 Score-blind complete-core construction

The two task compilers each produce 160 candidates per family. After collection, the release re-scores response syntax with one versioned parser and constructs a validity matrix. Within every panel-family stratum it selects the first 89 eligible task IDs under a fixed SHA-256 ordering. A task is eligible only when all 27 models completed it and produced a parseable three-item set. The selection code does not load the chosen ingredients or their scores until the task IDs are fixed. This yields a balanced common core without favoring any model or post-selecting easy tasks by observed score.

### 3.4 Uncertainty and multiplicity

All inferential choices are bound in the content-addressed analysis plan.

- **Scores.** 50000 bootstrap samples are drawn over ingredient anchors; tasks that share an anchor across panels always move together. Each resample is reduced to six panel-family means and then equally weighted. We report pointwise percentile intervals and a simultaneous max- $t$  band across all 27 model scores.
- **Pairs.** Every one of the 351 model pairs uses the same 534 tasks. Two-sided anchor-cluster sign-flip tests use 100000 Monte Carlo draws and Holm correction at familywise  $\alpha = .05$ . We also report paired mean differences, bootstrap intervals, and Cohen’s  $d_z$ .
- **Chance.** Each model is compared with the exact taskwise mean over all 56 portfolios. The 27 paired sign-flip tests form a separate Holm family.
- **Ranks.** Point ranks are accompanied by bootstrap rank intervals and contiguous statistical groups. Models that are not significantly separated are not presented as a resolved total order.

The design has 534 independent anchor clusters and 534 paired cells per contrast. Statistical claims are read directly from simultaneous intervals and Holm-adjusted tests; point-rank differences without corrected support are shown as unresolved.

## 4 Results

### 4.1 A statistically resolved FlavourBench leaderboard

All 27 endpoints receive a score on the same 534 tasks. Grok 4.6 has the largest point estimate, 65.1, with simultaneous 95% interval [61.0, 69.2]. Across all 351 prespecified paired contrasts, 101 remain significant after Holm correction. A point rank is therefore a compact summary, not a claim that every adjacent model is distinguishable. Figure 2 shows the score scale, simultaneous intervals, and rank groups; Table 3 gives exact values.

The ranked matrix is rectangular: every row contains 534 valid, parser-confirmed responses, so no endpoint is advantaged by answering a different subset. Provider failures and filtered calls remain available in the full execution ledger, but none appears as a zero-valued culinary

decision and no incomplete row enters this leaderboard.

## 4.2 Does the ordering replicate on a second task panel?

The two panel-specific scores use the same frozen metric and 89 tasks per family, but disjoint task IDs. Their model-level Pearson correlation is 0.89 and their rank correlation is 0.80. These are descriptive stability diagnostics, not a model-selection rule; the primary leaderboard and its uncertainty use both panels and cluster shared ingredient anchors. Figure 3 exposes every model rather than reducing replication to a single coefficient.

## 4.3 Which pairwise conclusions survive correction?

The score is continuous, and each model pair is compared on the same 534 tasks rather than on a small number of arena votes. Figure 4 visualizes the multiplicity-controlled results. Grey cells are scientifically useful: they mark pairs for which the shared task evidence does not support a directional claim at familywise  $\alpha = .05$ . This is different from declaring the two models equal.

## 4.4 Aggregate scores conceal different culinary profiles

Family-level results in Figure 5 show that similar aggregate scores can arise from different strengths. Constraint tasks penalize infeasible portfolios directly, while pairing and substitution reward graded semantic structure. The three columns are related but not interchangeable.

## 4.5 Rank uncertainty is visible, not hidden

Bootstrap rank intervals expose how often small score differences change order under plausible resampling of ingredient anchors. Figure 6 separates stable regions of the table from visually tempting but statistically unsupported micro-ranks. This is stricter than sorting point estimates and printing an ordinal list without uncertainty.

## 4.6 Real prompts and model decisions

Table 4 reports one deterministically selected common-core task from each family with responses from the highest- and lowest-point-estimate endpoints. Selection follows a fixed hash rule rather than task score. The public case-study JSON contains the complete prompt, all eight choices, the optimal portfolio, both raw model responses, selected ingredient identities, and exact scores. The examples illustrate why continuous scoring is preferable to exact match: two non-optimal portfolios can differ substantially in culinary utility.

# 5 From benchmark to training signal

The exhaustive task map makes FLAVOURBENCH useful beyond a static leaderboard. For a task state  $x_t$  and selected portfolio  $a$ , EPICURE supplies a deterministic reward  $r(x_t, a) = s_t(a)$ . This creates three immediate training interfaces:

1. supervised learning from the optimal portfolio and the full ordering of 56 candidates;
2. preference or contrastive learning from any pair  $(a_i, a_j)$  with known reward difference;
3. offline or online policy optimization using dense bounded rewards and held-out anchors.

The present experiment evaluates models rather than training them. The release supplies the environment, reward function, and anchor identifiers required for a clean next experiment: train on development reward maps and measure transfer on disjoint held-out anchors. Direct optimization on the public test map would measure memorization rather than culinary learning.

# 6 Reproducibility

The release is organized around content-addressed inputs and append-only outputs. It includes:

- both exact content-addressed task sets (534 scheduled tasks and 534 unique anchors), with prompts, candidates, all 56 portfolio scores, oracle provenance, task strata, and task-set digests;
- the 27-model manifest, exact provider route per model, fallback policy, supported parameters, endpoint contract, and route digest;
- all response records used to construct the 14418-cell ranked matrix, including raw text, completion state, usage, latency, parser output, task score, and digest;
- all content-addressed provider-attempt events referenced by those responses, including normalized and native finish reasons used to distinguish refusals from ordinary completions;
- the frozen joint analysis plan, anchor-cluster contract, seeds, 50,000 bootstrap and 100,000 sign-flip counts, all 351 pairwise rows, cross-panel diagnostics, and the final statistical release; and
- a verifier and asset builder that reconstruct the leaderboard CSV, pairwise CSV, LaTeX tables, and vector figures from local files without provider access.

Responses are written once under a model slot and cell identifier. Existing valid cells are skipped on resume; conflicting artifacts fail closed. Provider fallback is disabled, so the route shown in the manifest is the route measured. The task maps and analysis plan were frozen before the clean primary lineage; transport-calibration responses are retained separately and never enter the leaderboard.

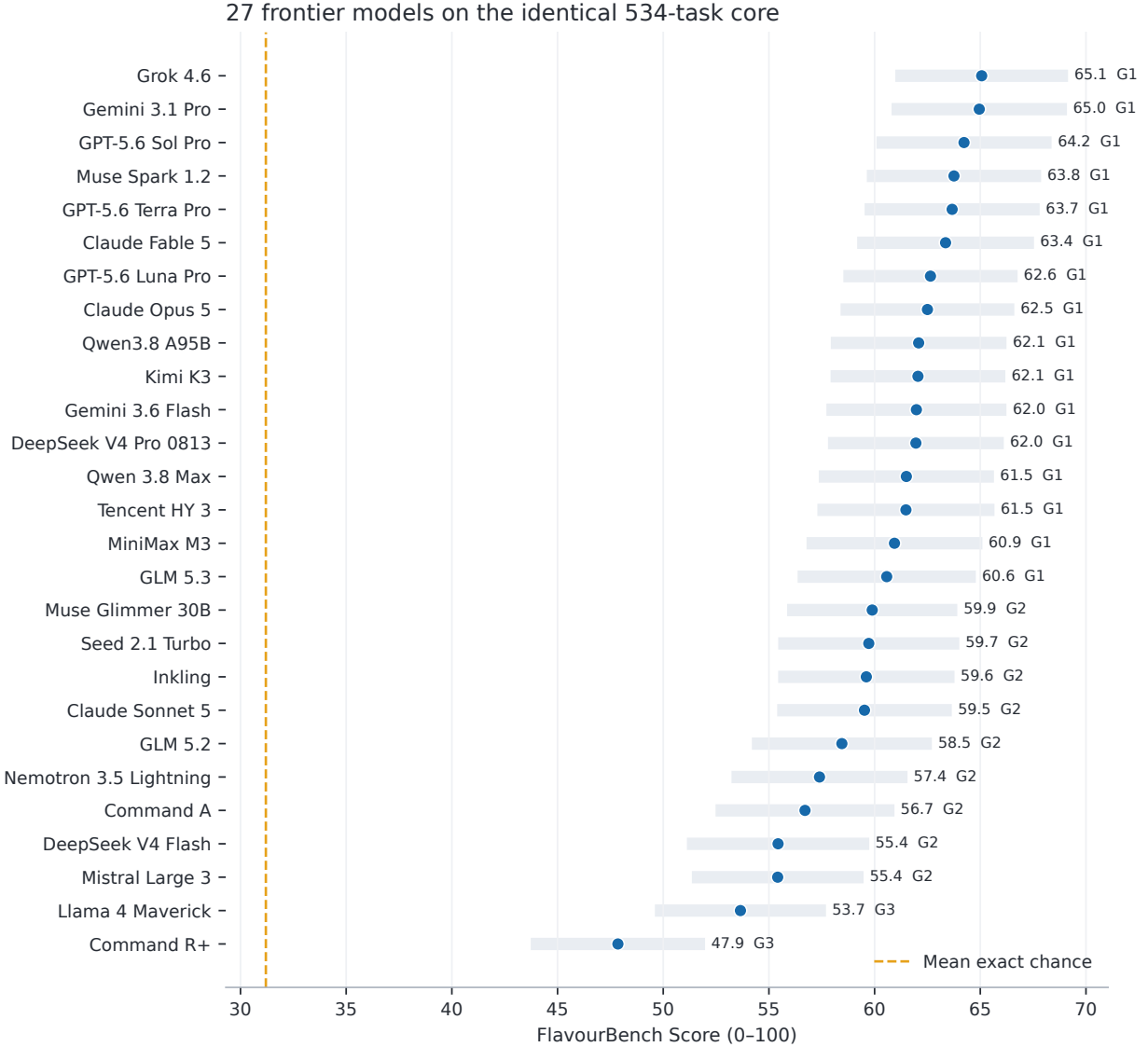


Figure 2: FlavourBench Score with simultaneous 95% max- $t$  intervals. The letter-free group numbers are inferential tiers: a point rank is shown for navigation, while models inside an unresolved group should not be read as a statistically established ordering.

## 7 Related work

General-purpose evaluation spans broad suites such as HELM [10], difficult knowledge tests such as MMLU-Pro and GPQA [16, 14], live contamination-resistant sets [17, 7], and human preference systems such as Chatbot Arena [2]. Model judges increase scale but introduce their own bias and calibration problems [20]. FLAVOURBENCH instead takes the executable- environment route used by software and agent benchmarks: SWE-bench runs repository tests [8], BFCL evaluates function calls [11], and  $\tau$ -bench checks interaction outcomes against database state [19].

Culinary datasets and benchmarks cover ingredient networks, recipe understanding, planning, nutrition, and food knowledge [1, 5, 15, 3, 18, 6, 4, 9]. Our focus is nar-

rower and complementary: common-task frontier model measurement against a versioned culinary decision environment, with exhaustive partial-credit maps and paired inference.

## 8 Limitations

FlavourBench Score measures agreement with one published EPICURE release, not universal human taste. Within-task min-max normalization makes the 0–100 scale comparable while discarding absolute raw-utility magnitudes. The tasks evaluate constrained ingredient selection rather than full recipe generation, sensory execution, safety advice, or long-horizon kitchen planning. The common-core requirement removes differential miss-



| Rank | Model                  | FB Score | simultaneous 95% CI | rank 95% CI | group |
|------|------------------------|----------|---------------------|-------------|-------|
| 1    | Grok 4.6               | 65.1     | [61.0, 69.2]        | [1, 5]      | 1     |
| 2    | Gemini 3.1 Pro         | 65.0     | [60.8, 69.1]        | [1, 6]      | 1     |
| 3    | GPT-5.6 Sol Pro        | 64.2     | [60.1, 68.4]        | [1, 8]      | 1     |
| 4    | Muse Spark 1.2         | 63.8     | [59.6, 67.9]        | [1, 10]     | 1     |
| 5    | GPT-5.6 Terra Pro      | 63.7     | [59.5, 67.8]        | [1, 11]     | 1     |
| 6    | Claude Fable 5         | 63.4     | [59.2, 67.5]        | [2, 13]     | 1     |
| 7    | GPT-5.6 Luna Pro       | 62.6     | [58.5, 66.8]        | [4, 14]     | 1     |
| 8    | Claude Opus 5          | 62.5     | [58.4, 66.6]        | [3, 15]     | 1     |
| 9    | Qwen3.8 A95B           | 62.1     | [57.9, 66.2]        | [4, 17]     | 1     |
| 10   | Kimi K3                | 62.1     | [57.9, 66.2]        | [5, 16]     | 1     |
| 11   | Gemini 3.6 Flash       | 62.0     | [57.7, 66.2]        | [4, 17]     | 1     |
| 12   | DeepSeek V4 Pro 0813   | 62.0     | [57.8, 66.1]        | [5, 17]     | 1     |
| 13   | Qwen 3.8 Max           | 61.5     | [57.4, 65.6]        | [7, 18]     | 1     |
| 14   | Tencent HY 3           | 61.5     | [57.3, 65.7]        | [6, 19]     | 1     |
| 15   | MiniMax M3             | 60.9     | [56.8, 65.1]        | [8, 20]     | 1     |
| 16   | GLM 5.3                | 60.6     | [56.4, 64.8]        | [8, 20]     | 1     |
| 17   | Muse Glimmer 30B       | 59.9     | [55.9, 63.9]        | [12, 21]    | 2     |
| 18   | Seed 2.1 Turbo         | 59.7     | [55.4, 64.0]        | [12, 22]    | 2     |
| 19   | Inkling                | 59.6     | [55.4, 63.8]        | [12, 22]    | 2     |
| 20   | Claude Sonnet 5        | 59.5     | [55.4, 63.7]        | [13, 22]    | 2     |
| 21   | GLM 5.2                | 58.5     | [54.2, 62.7]        | [16, 23]    | 2     |
| 22   | Nemotron 3.5 Lightning | 57.4     | [53.2, 61.6]        | [19, 25]    | 2     |
| 23   | Command A              | 56.7     | [52.5, 60.9]        | [20, 25]    | 2     |
| 24   | DeepSeek V4 Flash      | 55.4     | [51.1, 59.7]        | [22, 26]    | 2     |
| 25   | Mistral Large 3        | 55.4     | [51.4, 59.5]        | [22, 26]    | 2     |
| 26   | Llama 4 Maverick       | 53.7     | [49.6, 57.7]        | [24, 26]    | 3     |
| 27   | Command R+             | 47.9     | [43.7, 52.0]        | [27, 27]    | 3     |

Table 3: The complete automated leaderboard. Every score and every pairwise comparison uses the identical 534-task core. Rank intervals come from the anchor-cluster bootstrap; groups summarize the Holm-controlled pairwise graph.

| Family       | Prompt                                                                                                                                                                     | Higher-ranked selection                         | Lower-ranked selection                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------|
| Substitution | Select three alternatives to ‘boursin cheese’ that best preserve its dairy role, regional context, and mutual portfolio coherence.                                         | caciocavallo, fromage blanc, grana padano (100) | fromage blanc, quail egg, goose egg (15)    |
| Pairing      | Select the three-ingredient bundle with the strongest learned pairing to ‘sweetbread’ and the best internal coherence.                                                     | parsnip, cipollini onion, porcini mushroom (92) | parsnip, pickled onion, peppadew pepper (0) |
| Constraints  | For a dish centered on ‘potato’, select three vegetarian complements with NOVA processing level at most 2; among valid portfolios, maximize learned pairing and coherence. | parsley, black pepper, thyme (100)              | cheese, bay leaf, parsley (0)               |

Table 4: Three common-core response examples. Parentheses give the fixed task score. The complete prompts and unabridged responses are released in machine-readable form.

ingness from ranking, but it also restricts the estimand to tasks completed by every endpoint; the released supplemental tracks retain the rest. Results bind particular model routes and collection dates. The dense reward surface supports learning experiments, but this paper does not test whether optimizing EPICURE reward transfers to human cooking outcomes.

## 9 Conclusion

FLAVOURBENCH makes culinary reasoning measurable without asking one language model to grade another. A versioned runtime enumerates every candidate decision, a 27-model complete panel produces one continuous score, and paired uncertainty distinguishes resolved differences from apparent rank order. The result is an automated leaderboard that can be rerun, audited, and used as a dense training environment. Under this executable culinary ground truth and the released 534-task, 534-cluster distribution, every score, interval, contrast, and figure is

reconstructable from public artifacts.

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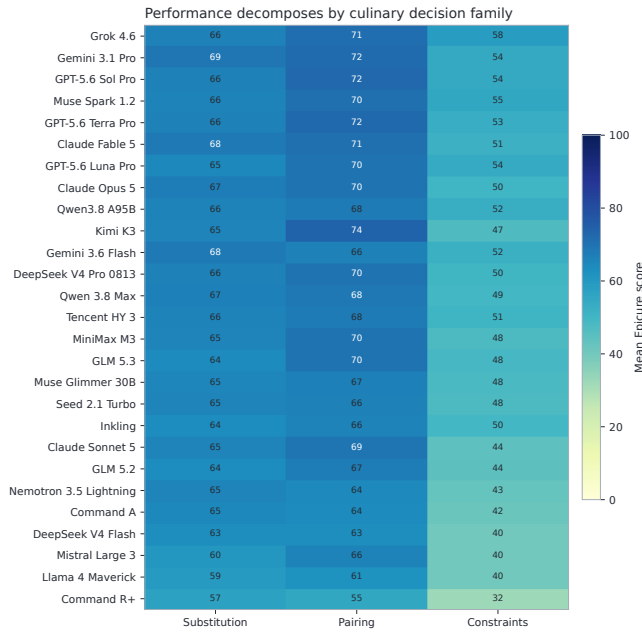


Figure 5: Mean score by family on the common core. Each family contributes 178 tasks per model. Values share the 0–100 scale but retain family-specific decision semantics.

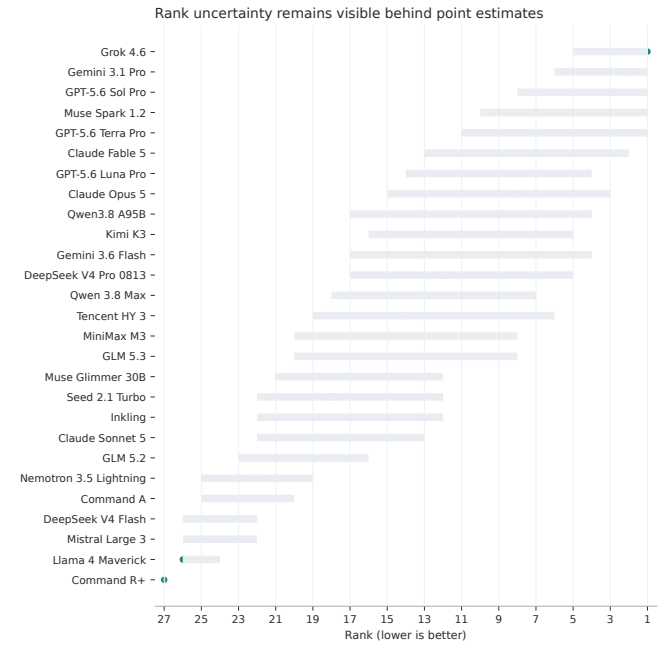


Figure 6: Point rank and 95% bootstrap rank interval for every endpoint. Narrow intervals imply a stable location; overlapping intervals mark unresolved local order.

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## A Exact execution routes

| Model                  | Backend           | Panel 1 route                | Panel 2 route                |
|------------------------|-------------------|------------------------------|------------------------------|
| Grok 4.6               | openrouter        | xai/zdr                      | xai/zdr                      |
| Gemini 3.1 Pro         | openrouter        | google-ai-studio             | google-ai-studio             |
| GPT-5.6 Sol Pro        | openrouter        | openai/flex                  | openai/flex                  |
| Muse Spark 1.2         | openrouter        | meta                         | meta                         |
| GPT-5.6 Terra Pro      | openrouter        | openai/flex                  | openai/flex                  |
| Claude Fable 5         | openrouter        | google-vertex/global         | google-vertex/global         |
| GPT-5.6 Luna Pro       | openrouter        | openai/flex                  | openai                       |
| Claude Opus 5          | openrouter        | azure/us                     | azure/us                     |
| Qwen3.8 A95B           | openrouter        | alibaba                      | alibaba                      |
| Kimi K3                | openrouter        | morph                        | morph                        |
| Gemini 3.6 Flash       | openrouter        | google-vertex/us             | google-vertex/us             |
| DeepSeek V4 Pro 0813   | openrouter        | gmcloud/fp8                  | gmcloud/fp8                  |
| Qwen 3.8 Max           | openrouter        | alibaba                      | alibaba                      |
| Tencent HY 3           | openrouter        | gmcloud/bf16                 | gmcloud/bf16                 |
| MiniMax M3             | openrouter        | venice/fp8                   | venice/fp8                   |
| GLM 5.3                | zai_coding_direct | zai-coding-plan-direct       | zai-coding-plan-direct       |
| Muse Glimmer 30B       | openrouter        | fireworks                    | fireworks                    |
| Seed 2.1 Turbo         | openrouter        | seed/fp8                     | seed/fp8                     |
| Inkling                | openrouter        | deepinfra/fp8                | deepinfra/fp8                |
| Claude Sonnet 5        | openrouter        | amazon-bedrock/claude-on-aws | amazon-bedrock/claude-on-aws |
| GLM 5.2                | openrouter        | coreweave/fp4                | coreweave/fp4                |
| Nemotron 3.5 Lightning | openrouter        | venice/fp4                   | venice/fp4                   |
| Command A              | openrouter        | cohere                       | cohere                       |
| DeepSeek V4 Flash      | openrouter        | deepinfra/fp4                | deepinfra/fp8                |
| Mistral Large 3        | openrouter        | mistral                      | mistral                      |
| Llama 4 Maverick       | openrouter        | digitalocean                 | digitalocean                 |
| Command R+             | openrouter        | cohere                       | cohere                       |

Table 5: Frozen execution routes by panel. Automatic fallback is disabled; route differences indicate complete, score-blind model-block replacements rather than cell pooling.

## B Per-family scores

| Model                  | Substitution | Pairing | Constraints |
|------------------------|--------------|---------|-------------|
| Grok 4.6               | 66.1         | 70.9    | 58.2        |
| Gemini 3.1 Pro         | 69.0         | 71.6    | 54.2        |
| GPT-5.6 Sol Pro        | 66.2         | 72.5    | 54.1        |
| Muse Spark 1.2         | 65.9         | 70.3    | 55.0        |
| GPT-5.6 Terra Pro      | 66.1         | 72.1    | 52.8        |
| Claude Fable 5         | 68.2         | 70.6    | 51.3        |
| GPT-5.6 Luna Pro       | 64.5         | 69.7    | 53.7        |
| Claude Opus 5          | 67.2         | 69.9    | 50.4        |
| Qwen3.8 A95B           | 66.0         | 67.8    | 52.5        |
| Kimi K3                | 65.5         | 73.7    | 47.0        |
| Gemini 3.6 Flash       | 68.1         | 66.2    | 51.6        |
| DeepSeek V4 Pro 0813   | 66.4         | 69.5    | 49.9        |
| Qwen 3.8 Max           | 66.6         | 68.5    | 49.5        |
| Tencent HY 3           | 66.0         | 67.9    | 50.5        |
| MiniMax M3             | 65.4         | 69.9    | 47.6        |
| GLM 5.3                | 63.7         | 69.6    | 48.4        |
| Muse Glimmer 30B       | 65.1         | 66.8    | 47.8        |
| Seed 2.1 Turbo         | 65.3         | 66.2    | 47.6        |
| Inkling                | 63.6         | 65.8    | 49.5        |
| Claude Sonnet 5        | 65.3         | 69.2    | 44.1        |
| GLM 5.2                | 63.6         | 67.2    | 44.5        |
| Nemotron 3.5 Lightning | 65.3         | 64.3    | 42.5        |
| Command A              | 64.7         | 63.9    | 41.6        |
| DeepSeek V4 Flash      | 63.1         | 63.2    | 40.1        |
| Mistral Large 3        | 60.4         | 65.8    | 40.0        |
| Llama 4 Maverick       | 59.4         | 61.4    | 40.2        |
| Command R+             | 56.7         | 54.9    | 32.0        |

Table 6: Mean score in each 178-task family. Every entry uses the same tasks for every model; the equal-family mean is the FlavourBench Score.

# C Protocol constants

| Quantity                   | Frozen value                    |
|----------------------------|---------------------------------|
| Models                     | 27                              |
| Collection panels          | 2                               |
| Primary tasks per model    | 534                             |
| Unique anchor clusters     | 534                             |
| Tasks per family           | 178                             |
| Candidate ingredients      | 8                               |
| Selected ingredients       | 3                               |
| Scored portfolios per task | 56                              |
| Complete response cells    | 14418                           |
| Bootstrap replicates       | 50000                           |
| Sign-flip replicates       | 100000                          |
| Pairwise hypotheses        | 351                             |
| Shared cells per pair      | 534                             |
| Common-core rule           | Status/parser only; score-blind |

Table 7: Core design and inference constants fixed before clean primary execution.